

Surface Metrics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Text Descriptives on the Exchange-Aligned Structure

PENPAL Project

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1 Overview

This notebook compares surface-level text metrics across conditions using the long-format, exchange-aligned metric tables.

The underlying processed data are now one row per **speaker contribution**, with recovered role metadata:

- **Human / AI** in Human-AI
- **Starter-side / Non-starter-side** in Human-Human and AI-AI

Those role labels are useful for **within-condition description**, but they are not the primary cross-condition comparison axis. For cross-condition comparison, this notebook uses **Slot 1 / Slot 2 asymmetry**.

The notebook therefore no longer assumes old wide `author_1_*` / `author_2_*` columns.

Primary cross-condition object. Per story and per metric m ,

$$A_m(i) = |\overline{m}_{\text{slot}=1, i} - \overline{m}_{\text{slot}=2, i}|.$$

This is the unsigned, label-invariant side-difference magnitude. For each metric we test $A_m \sim \text{condition}$ with ANOVA + Kruskal and apply BH adjustment across the metric panel. The LMM slot-marginal contrasts are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded surface-metric rows by condition

condition	n_stories	n_rows
Human-AI	100	1708
Human-Human	36	648
AI-AI	40	720

3 Metric Selection

Table 2: Surface metrics selected for the comparison notebook

metric_alias	column_name
word_count	n_tokens
sentence_length	sentence_length_mean
lexical_diversity	proportion_unique_tokens
readability	flesch_reading_ease
coherence	first_order_coherence

4 Primary: Per-Story Side-Mean Difference

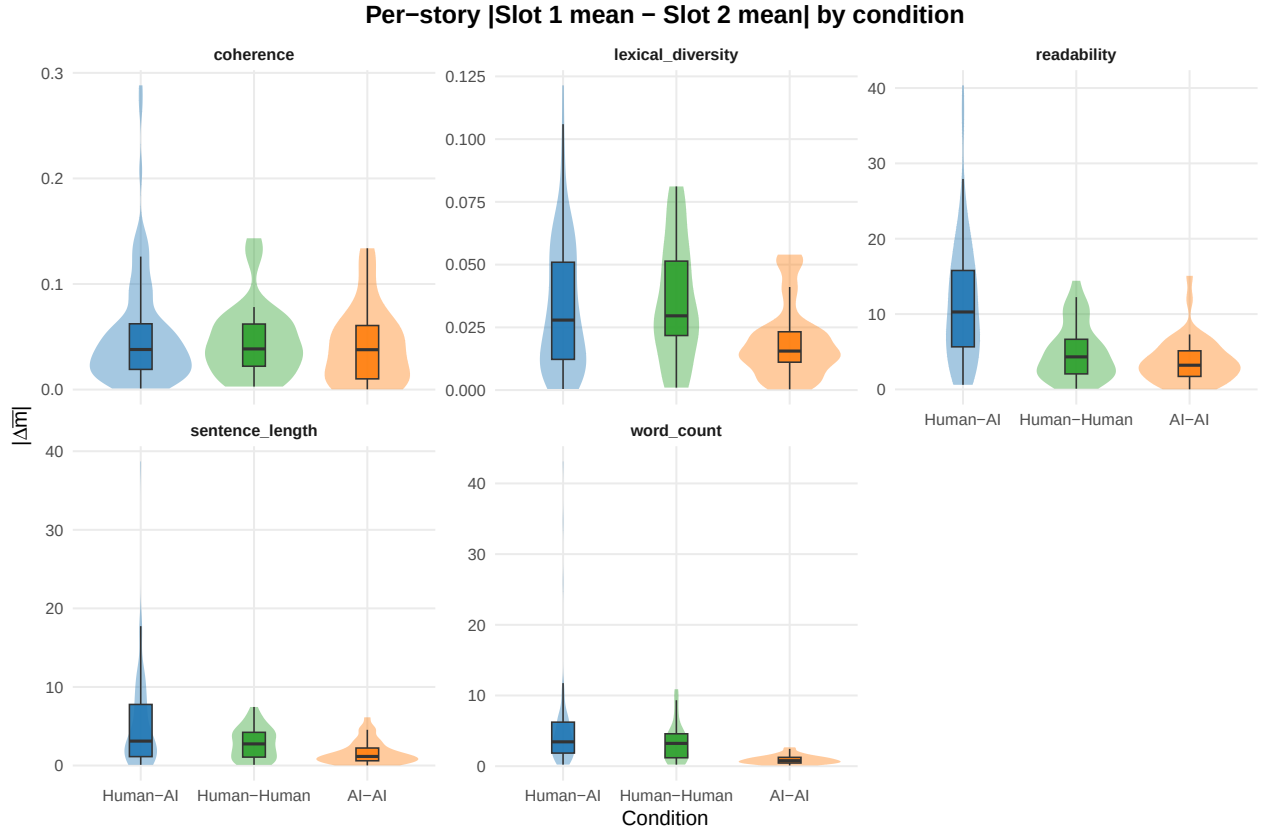
Table 3: Per-story |side-mean difference| by condition, per metric

metric	condition	n	mean_A	median_A	sd_A
coherence	Human-AI	98	0.053	0.038	0.052
coherence	Human-Human	36	0.046	0.039	0.035
coherence	AI-AI	39	0.042	0.038	0.037
lexical_diversity	Human-AI	100	0.033	0.028	0.026
lexical_diversity	Human-Human	36	0.035	0.030	0.024
lexical_diversity	AI-AI	40	0.019	0.016	0.015
readability	Human-AI	100	11.447	10.277	8.084
readability	Human-Human	36	4.928	4.320	3.727
readability	AI-AI	40	3.793	3.210	3.086
sentence_length	Human-AI	100	5.154	3.111	5.658
sentence_length	Human-Human	36	2.882	2.755	2.019
sentence_length	AI-AI	40	1.661	1.157	1.505
word_count	Human-AI	100	4.822	3.444	6.019
word_count	Human-Human	36	3.497	3.222	2.792
word_count	AI-AI	40	0.944	0.778	0.620

Table 4: Primary cross-condition tests for $A_m \sim \text{condition}$, per metric (BH-adjusted across metrics)

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
coherence	0.4242	0.6109	0.4242	0.6109
lexical_diversity	0.0032	0.0042	0.0040	0.0052

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
readability	0.0000	0.0000	0.0000	0.0000
sentence_length	0.0001	0.0001	0.0002	0.0001
word_count	0.0001	0.0000	0.0002	0.0000

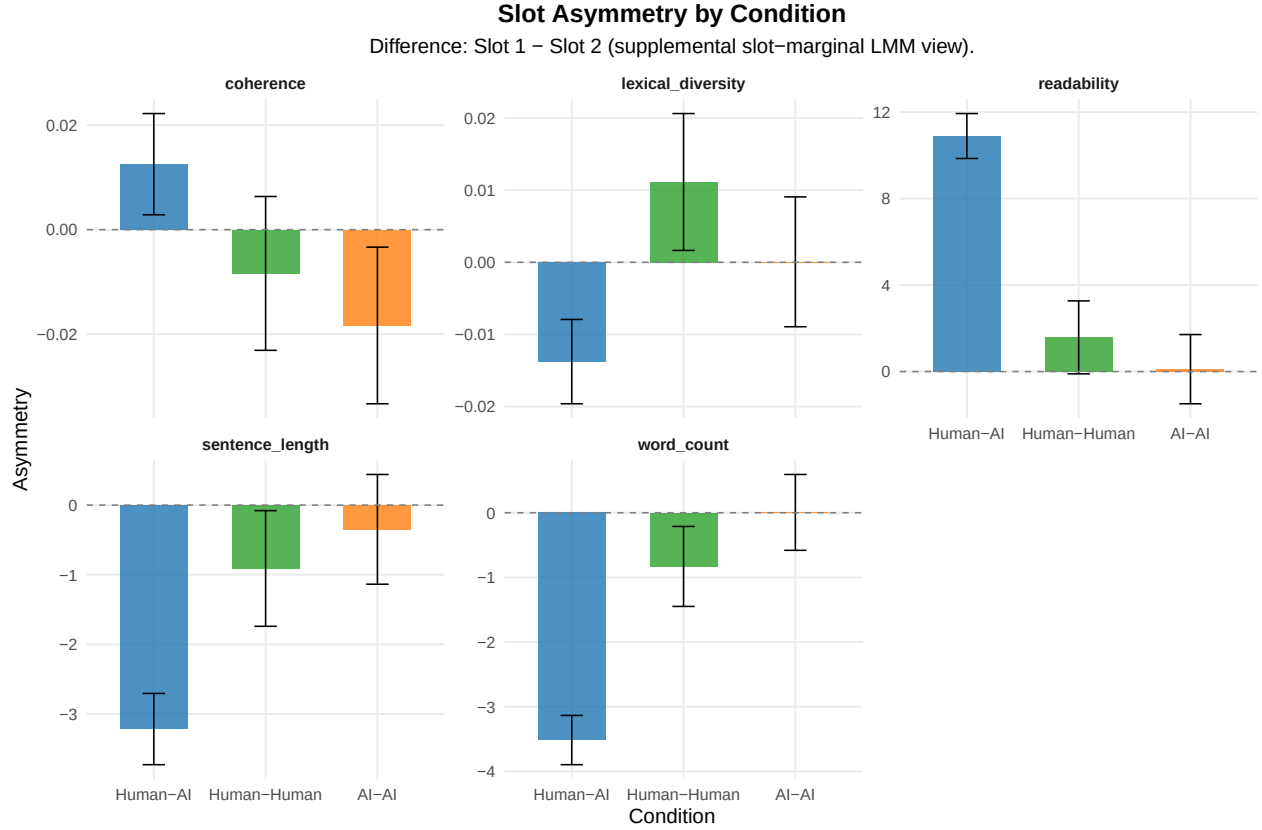


5 Supplemental: Slot-Marginal LMM per Metric

Table 5: Slot asymmetry by condition for selected surface metrics

metric	condition	slot_1_mean	slot_2_mean	difference	se	p	t
word_count	Human-AI	25.581	29.096	-3.515	0.194	0.000	NA
word_count	Human-Human	26.031	26.861	-0.830	0.316	0.009	NA
word_count	AI-AI	27.972	27.967	0.006	0.299	0.985	NA
sentence_length	Human-AI	11.577	14.794	-3.217	0.261	0.000	NA
sentence_length	Human-Human	10.329	11.239	-0.910	0.424	0.032	NA
sentence_length	AI-AI	14.904	15.252	-0.348	0.402	0.387	NA
lexical_diversity	Human-AI	0.871	0.885	-0.014	0.003	0.000	NA
lexical_diversity	Human-Human	0.871	0.860	0.011	0.005	0.021	NA
lexical_diversity	AI-AI	0.879	0.879	0.000	0.005	0.988	NA
readability	Human-AI	90.727	79.835	10.892	0.530	0.000	NA

metric	condition	slot_1_mean	slot_2_mean	difference	se	p	t
readability	Human-Human	93.714	92.134	1.581	0.861	0.066	NA
readability	AI-AI	83.061	82.951	0.110	0.817	0.892	NA
coherence	Human-AI	0.803	0.790	0.013	0.005	0.011	2.539
coherence	Human-Human	0.811	0.819	-0.008	0.008	0.265	-1.114
coherence	AI-AI	0.777	0.795	-0.018	0.008	0.017	-2.396

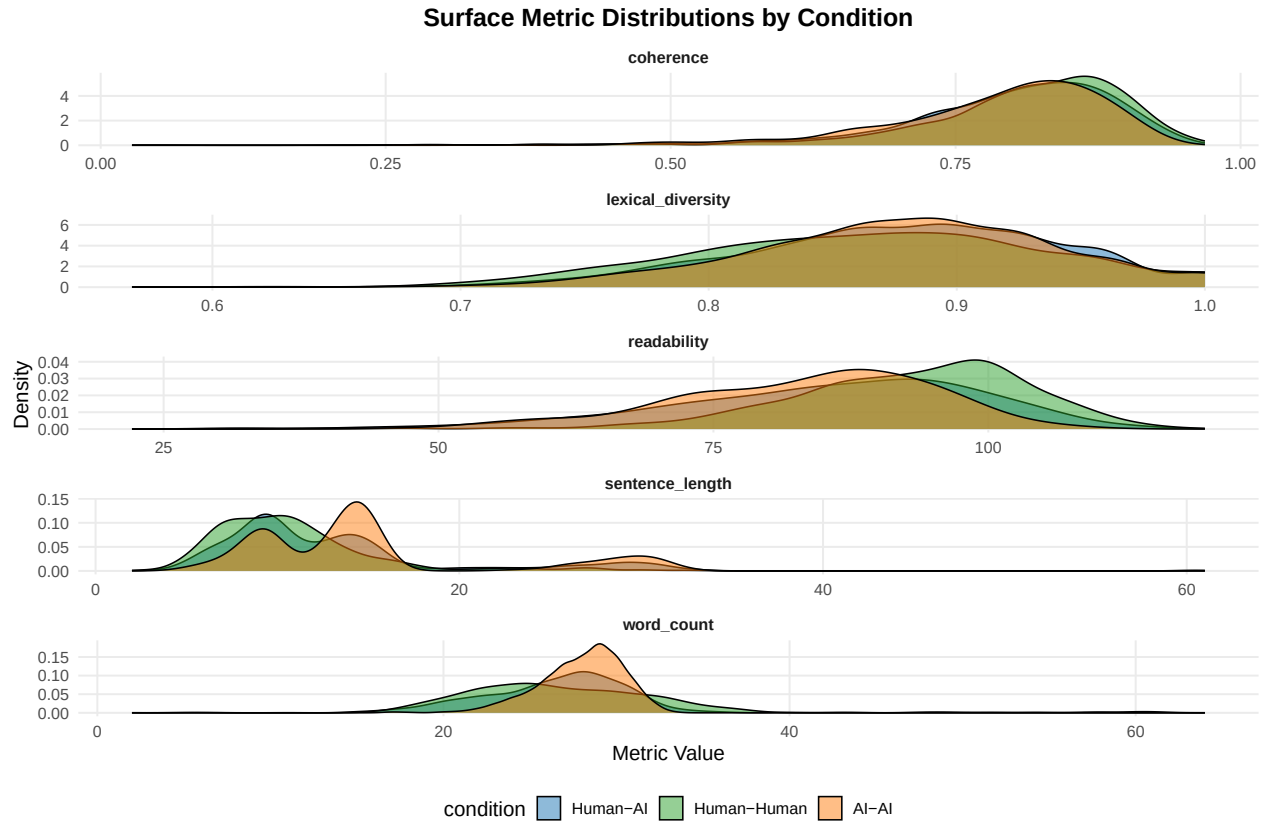


6 Supplemental: General Condition-Level Comparison

Table 6: Condition-level descriptives for selected surface metrics

condition	metric	mean_value	sd_value
Human-AI	word_count	27.320	6.430
Human-Human	word_count	26.446	4.767
AI-AI	word_count	27.969	2.409
Human-AI	sentence_length	13.231	7.108
Human-Human	sentence_length	10.784	4.544
AI-AI	sentence_length	15.078	6.987
Human-AI	lexical_diversity	0.878	0.065
Human-Human	lexical_diversity	0.866	0.071
AI-AI	lexical_diversity	0.879	0.061
Human-AI	readability	85.200	14.454

condition	metric	mean_value	sd_value
Human-Human	readability	92.924	10.656
AI-AI	readability	83.006	12.020
Human-AI	coherence	0.797	0.101
Human-Human	coherence	0.815	0.085
AI-AI	coherence	0.785	0.099



```
##
## === word_count ( n_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 18056
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.8933 -0.5130  0.0762  0.5305  5.4753
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 11.90      3.449
## Residual                17.97      4.239
## Number of obs: 3076, groups: conversation_id, 176
##
```

```

## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      27.3384    0.3599 172.1337  75.963   <2e-16 ***
## conditionHuman-Human -0.8924    0.6984 171.0191  -1.278    0.203
## conditionAI-AI       0.6310    0.6722 171.0509   0.939    0.349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.515
## condtnAI-AI -0.535  0.276
##
## contrast               estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human)   0.892 0.698 Inf   1.278  0.6039
## (Human-AI) - (AI-AI)        -0.631 0.672 Inf  -0.939  1.0000
## (Human-Human) - (AI-AI)     -1.523 0.825 Inf  -1.847  0.1944
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === sentence_length ( sentence_length_mean ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 19630.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.0166 -0.6038 -0.1440  0.3938  5.7967
##
## Random effects:
## Groups           Name      Variance Std.Dev.
## conversation_id (Intercept) 13.25    3.640
## Residual                 30.66    5.537
## Number of obs: 3076, groups: conversation_id, 176
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      13.1856    0.3879 174.5026  33.989   < 2e-16 ***
## conditionHuman-Human -2.4020    0.7522 172.8550  -3.193   0.00167 **
## conditionAI-AI       1.8923    0.7241 172.9020   2.613   0.00976 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.516
## condtnAI-AI -0.536  0.276
##
## contrast               estimate      SE  df z.ratio p.value
## (Human-AI) - (Human-Human)   2.40 0.752 Inf   3.193  0.0042
## (Human-AI) - (AI-AI)        -1.89 0.724 Inf  -2.613  0.0269

```

```

## (Human-Human) - (AI-AI)          -4.29 0.888 Inf  -4.834  <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === lexical_diversity ( proportion_unique_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: -8165.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5848 -0.6277  0.0539  0.6711  2.5984
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0004813 0.02194
## Residual                  0.0038210 0.06181
## Number of obs: 3076, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.878083   0.002656 175.584909 330.572  <2e-16 ***
## conditionHuman-Human -0.012563   0.005130 171.315689  -2.449   0.0153 *
## conditionAI-AI      0.001313   0.004939 171.437014   0.266   0.7907
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.278
##
## contrast            estimate      SE df z.ratio p.value
## (Human-AI) - (Human-Human)  0.01256 0.00513 Inf  2.449  0.0430
## (Human-AI) - (AI-AI)      -0.00131 0.00494 Inf  -0.266  1.0000
## (Human-Human) - (AI-AI)    -0.01388 0.00605 Inf  -2.293  0.0655
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === readability ( flesch_reading_ease ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 24172.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max

```

```

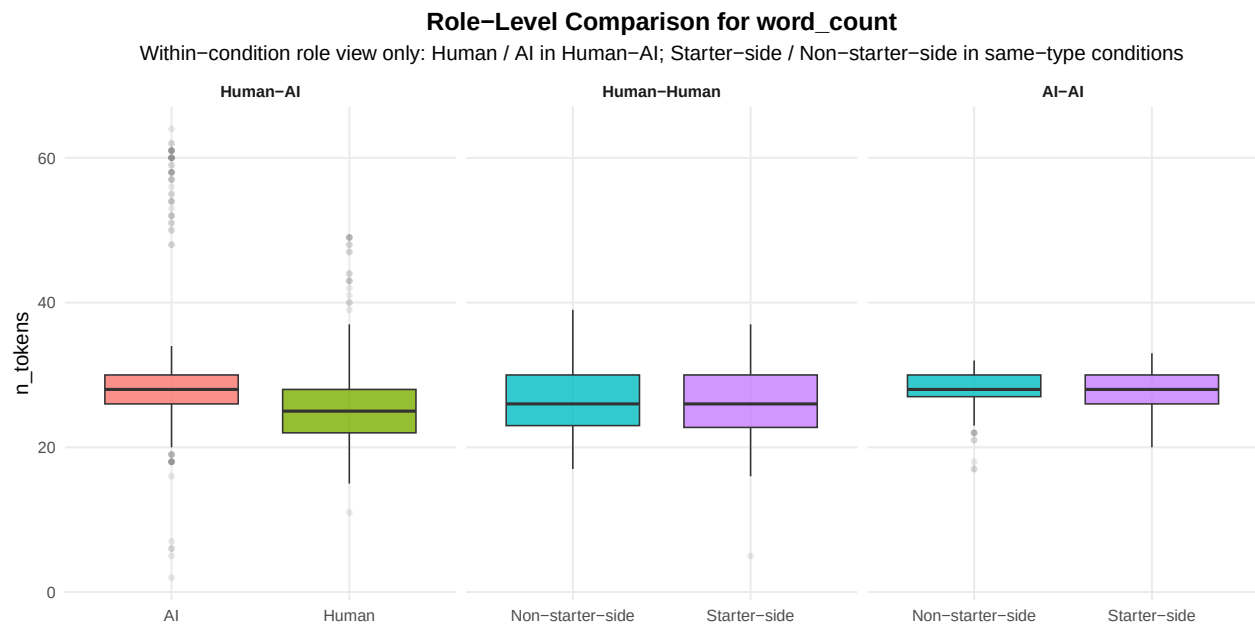
## -4.5598 -0.5983 0.0999 0.6781 2.8263
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 36.92 6.077
## Residual 137.51 11.727
## Number of obs: 3076, groups: conversation_id, 176
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 85.2791 0.6708 174.2882 127.125 < 2e-16 ***
## conditionHuman-Human 7.6448 1.2992 171.8427 5.884 2.04e-08 ***
## conditionAI-AI -2.2733 1.2507 171.9124 -1.818 0.0709 .
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) cndH-H
## cndtnHmn-Hm -0.516
## condtnAI-AI -0.536 0.277
##
## contrast estimate SE df z.ratio p.value
## (Human-AI) - (Human-Human) -7.64 1.30 Inf -5.884 <.0001
## (Human-AI) - (AI-AI) 2.27 1.25 Inf 1.818 0.2073
## (Human-Human) - (AI-AI) 9.92 1.53 Inf 6.467 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: bonferroni method for 3 tests
##
## === coherence ( first_order_coherence ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: -4884.9
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -7.6866 -0.4510 0.1641 0.6475 2.4876
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 0.0008456 0.02908
## Residual 0.0086135 0.09281
## Number of obs: 2643, groups: conversation_id, 176
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 0.797104 0.003823 174.859293 208.503 <2e-16 ***
## conditionHuman-Human 0.018006 0.007230 159.288188 2.491 0.0138 *
## conditionAI-AI -0.011408 0.007195 171.072787 -1.585 0.1147
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```



```
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H
## cndtnHmn-Hm -0.529
## condtnAI-AI -0.531  0.281
## contrast      estimate      SE  df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0180 0.00723 160  -2.491  0.0413
## (Human-AI) - (AI-AI)      0.0114 0.00720 172   1.585  0.3443
## (Human-Human) - (AI-AI)    0.0294 0.00865 162   3.400  0.0025
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Role-Level Comparison



```
##
## === Role contrast for word_count ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 10299
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.1037 -0.5035  0.0205  0.4846  4.6698
##
## Random effects:
## Groups      Name          Variance Std.Dev.
```

```

## conversation_id (Intercept) 18.50    4.302
## Residual                    20.68    4.548
## Number of obs: 1708, groups: conversation_id, 100
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      29.0964    0.4575  110.9134   63.60  <2e-16 ***
## condition_roleHuman -3.5152    0.2201 1606.4692  -15.97  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlHmn -0.241
## contrast      estimate    SE   df t.ratio p.value
## AI - Human      3.52 0.22 1607  15.972  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 3752.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.3416 -0.7182 -0.0558  0.7077  2.7770
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept)  5.572   2.361
## Residual                17.270   4.156
## Number of obs: 648, groups: conversation_id, 36
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      26.6728    0.4562  45.9806   58.47  <2e-16 ***
## condition_roleStarter-side -0.4537    0.3265 611.0000   -1.39    0.165
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.358
## contrast                        estimate    SE   df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.454 0.327 611   1.390  0.1652
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 3191.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9283 -0.5796  0.1691  0.6762  2.4627
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 1.419         1.191
## Residual                  4.424         2.103
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      28.00278    0.21854  51.30301 128.136 <2e-16 ***
## condition_roleStarter-side -0.06667    0.15678 679.00000  -0.425    0.671
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.359
## contrast              estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.0667 0.157 679   0.425  0.6708
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for sentence_length ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 11082.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5045 -0.6047 -0.1818  0.3729  5.2410
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 13.64         3.693
## Residual                  34.16         5.845
## Number of obs: 1708, groups: conversation_id, 100
##

```

```

## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      14.7946    0.4200  126.5688   35.22  <2e-16 ***
## condition_roleHuman -3.2169    0.2829 1607.6182  -11.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlHmn -0.337
## contrast      estimate    SE   df t.ratio p.value
## AI - Human      3.22 0.283 1607  11.373  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 3794.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.7525 -0.6499 -0.1814  0.3242  5.1000
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept)  0.886   0.9413
## Residual                19.811   4.4509
## Number of obs: 648, groups: conversation_id, 36
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      10.8563    0.2928  83.0638   37.073  <2e-16 ***
## condition_roleStarter-side -0.1453    0.3497 611.0000   -0.415    0.678
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.597
## contrast              estimate    SE   df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.145 0.35 611    0.415  0.6780
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))

```

```

## Data: df_subset
##
## REML criterion at convergence: 4487.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2291 -0.7053  0.0685  0.5080  3.4982
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 23.70      4.868
## Residual                  25.52      5.052
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      15.5219    0.8145  43.5195  19.057  <2e-16 ***
## condition_roleStarter-side -0.8880    0.3766 679.0000  -2.358  0.0186 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.231
## contrast              estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.888 0.377 679  2.358  0.0186
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for lexical_diversity ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: -4553.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9736 -0.6207  0.0545  0.6904  2.5734
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0004195 0.02048
## Residual                  0.0037849 0.06152
## Number of obs: 1708, groups: conversation_id, 100
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.850e-01  2.938e-03  1.774e+02 301.201  < 2e-16 ***
## condition_roleHuman -1.376e-02  2.977e-03  1.607e+03  -4.621  4.12e-06 ***

```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cndtn_rlHmn -0.507
## contrast      estimate      SE   df t.ratio p.value
## AI - Human    0.0138 0.00298 1607   4.621  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: -1584.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1625 -0.6663  0.0585  0.6867  2.1380
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.0003801 0.01950
## Residual              0.0047111 0.06864
## Number of obs: 648, groups:  conversation_id, 36
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.654e-01  5.010e-03 6.871e+01 172.734  <2e-16 ***
## condition_roleStarter-side 3.274e-04  5.393e-03 6.110e+02   0.061   0.952
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## cndtn_rlSt- -0.538
## contrast      estimate      SE   df t.ratio p.value
## (Non-starter-side) - (Starter-side) -0.000327 0.00539 611  -0.061  0.9516
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: -2055.2
##

```

```

## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.96440 -0.64985  0.03278  0.63825  2.75075
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0007296 0.02701
## Residual                    0.0030035 0.05480
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.785e-01  5.156e-03 5.476e+01  170.38  <2e-16 ***
## condition_roleStarter-side 1.798e-03  4.085e-03 6.790e+02   0.44    0.66
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.396
## contrast              estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side) -0.0018 0.00408 679  -0.440  0.6599
##
## Degrees-of-freedom method: kenward-roger

```

8 Summary

Primary cross-condition result: per-metric $A_m(i) = |\overline{m}_{\text{slot } 1, i} - \overline{m}_{\text{slot } 2, i}|$, tested with ANOVA + Kruskal and BH-adjusted across metrics.

Supplemental views retained:

- LMM slot-marginal contrasts per metric (`metric ~ slot * condition + (1|story)`)
- condition-level distributions and marginal condition LMMs
- within-condition role comparisons (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized side differences across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-26 20:17:04.120657